Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/11/2007
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Capillary Flow in Complex Geometries		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	05/04/2005	**End Date:**	05/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05AA29A		
Performance Goal No.:			
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Task Description:	Knowledge of capillary flows in complex geometries is fundamental to many space fluids systems designs. Such knowledge provides a basis to exploit passive means of containment and management for large liquid inventories aboard spacecraft. Such knowledge is also essential to prevent inadvertent destabilization and deactivation of critical systems where capillary forces can dominate fluid behavior. Complimentary but distinctly non-computationally-intensive methods to design and analysis such are in need by the aerospace community tasked to meet NASA's challenges for space exploration in the present and future. Specific progress is reported here for three fundamental and applied research projects: (1) the NRA investigation "Capillary Flow in Complex Geometries," (2) ISS Flight experiment results and analysis for the "Capillary Flow Experiment" (CFE), and (3) progress in the development of the German-US collaborative ISS experiment "Capillary Channel Flow" (CCF) experiment. All are briefly highlighted in this summary and addressed individually in greater detail in the subsequent report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the results of this work are immediately useful in the design of fluid systems aboard space craft. Certain results also inform the design and analysis methods for porous structures and media for the study of transport within porous media on earth such as ground water flows, high performance wicks for heat pipes, improved Lab-on Chip methods, and addressing flows in fuel cells.
Task Progress:	Specific progress is reported here for three fundamental and applied research projects: (1) the NRA investigation "Capillary Flow in Complex Geometries," (2) ISS Flight experiment results and analysis for the "Capillary Flow Experiment" (CFE), and (3) progress in the development of the German-US collaborative ISS experiment "Capillary Channel Flow" (CCF) experiment. All are briefly highlighted in this summary and addressed individually in greater detail in the subsequent report. 1. NRA-Research, Capillary Flow in Complex Geometries. This work has produced numerous design guides for passive fluid control aboard spacecraft. The work has also spun out methods to address microscale capillary flows in porous materials for terrestrial applications. Significant progress in the first year research effort was made. For example, a first thorough analysis for flows in rounded interior was completed and published in the Journal of Fluid Mechanics. Drop tower experiments performed at NASA GRC were conducted to benchmark the theoretical predictions. Thirty-five drop tower experiments were performed to investigate flows in 3-D corner networks, containers with interior vanes, and corners possessing gaps—models of systems commonly employed in spacecraft propellant management schemes. Experiments on the shapes and stability of large wall-edge-bound capillary drops and bubbles were computed and similar computations are underway for drops and bubbles on 3-D structures such as screens also employed in low-g fluid management applications. One journal articles appeared in the literature concerning this research and another is currently in preparation. Three conference papers appeared and a technical report was published as part of a larger NASA technical memorandum. Five technical conference presentations were made, with at least 13 outreach opportunities exploited to introduce local area students to low-g physical phenomena, NASA-related research, ISS utilization, and the challenges of aerospace engineering. Two M.S.M.E. students, one Post Doctoral Fellow, one visiting scholar (Bremen, Germany), three summer high school students, and over 18 undergraduate students were in part supported by this research effort. 2. Capillary Flow Experiment (CFE): The CFE was operated on 7 separate occasions on the ISS during this period by Astronauts Jeff Williams, Sunita Williams, and Michael Lopez-Alegria. The objective of the CFE experiments are to determine the impact of certain capillary phenomena critical to the management of liquid inventories aboard space craft. These in clued the moving contact line boundary condition, critical geometric wetting conditions and complex 3-D wicking. All of the experiments were 100% successful with many tests providing additional science helpful in extending current design capabilities. For certain experiments such as CFE-CL, many hundreds of data points were collected for comparison and guidance of predictions. Other images recorded for other experiments will go directly into journal articles in preparation. Five more CFE tests aboard ISS are planned. This work has in part supported two M.S.M.E. students, one undergraduate student, three summer high school students, and one visiting scholar (Bremen, Germany). 3. Capillary Channel Flow (CCF): In this work a collaboration between German investigator Michael Dreyer (Univ. Bremen/ZARM) and U.S. investigator M. Weislogel (Portland State University) was established to guide the German hardware design and development for future experiments aboard the ISS (Glovebox). The experiments concern the performance and stability of forced capillary flows and their dependence on system geometry. Such flows are critical to advanced low-gravity fuels and cryogens management. The science requirements for the hardware to meet the US objectives were completed during this period. Anticipated space flight of CCF is expected several years from now.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Chen Y, Weislogel MM, Nardin C. "Capillary-driven flows along rounded interior corners." J. Fluid Mech. 2006 Nov;566:235-71. http://dx.doi.org/10.1017/S0022112006001996 , Nov-2006
Papers from Meeting Proceedings	Weislogel MM, Jenson R, Klatte J, Dreyer M. "Interim Results from the Capillary Flow Experiment Aboard ISS: The Moving Contact Line Boundary Condition." 45th AIAA Aerospace Sci. Meeting and Exhibit, Reno, Nevada, Jan. 8-11, 2007. AIAA-2007-747 paper, January 2007. , Jan-2007
Papers from Meeting Proceedings	Chen Y, Bolleddula D, Weislogel M. "Capillary Flow in Cylindrical Containers with Rounded Interior Corners." 45th AIAA Aerospace Sci. Meeting and Exhibit, Reno, Nevada, Jan. 8-11, 2007. AIAA-2007-167 paper, January 2007. , Jan-2007
Papers from Meeting Proceedings	Weislogel M, Jenson R, Bolleddula D. "Capillary Driven Flows in Weakly 3-Dimensional Polygonal Containers." 45th AIAA Aerospace Sci. Meeting and Exhibit, Reno, Nevada, Jan. 8-11, 2007. AIAA-2007-748 paper, January 2007. , Jan-2007